

SHABBONA 5 NNE, IL, USA

WMO: 724970

Lat: 41.8430N

Lon: 88.8513W

Elev: 861

StdP: 14.24

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 54811

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.8	-1.8	-13.4	2.7	-7.0	-8.5	3.6	-1.3	24.2	20.9	21.7	16.8	7.8	N/A	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	88.7	73.5	86.2	72.6	83.7	71.5	77.3	85.4	75.3	82.8	73.6	81.3	5.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.7	134.3	82.8	72.9	126.2	80.7	71.0	118.1	78.9	41.3	85.5	39.3	82.6	37.8	81.6	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.1	17.2	15.1	-14.2	92.2	6.8	2.6	-19.1	94.1	-23.1	95.6	-26.9	97.0	-31.9	98.9		
			-14.2	80.0	7.3	1.4	-19.5	81.0	-23.8	81.8	-27.9	82.6	-33.2	83.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	20.9	23.4	36.8	48.2	59.8	69.0	71.0	68.8	63.1	50.9	38.0	26.3
	DBStd	20.12	12.50	12.13	11.22	9.08	8.94	6.05	5.67	5.23	7.71	9.30	9.78	11.22
	HDD50	3458	904	745	435	140	16	0	0	0	4	103	375	736
	HDD65	6775	1369	1164	876	507	212	28	13	23	125	447	811	1200
	CDD50	2779	0	1	25	86	321	570	651	582	396	132	14	1
	CDD65	621	0	0	1	3	52	148	199	140	68	10	0	0
	CDH74	5634	0	0	13	63	533	1317	1791	1147	657	113	0	0
	CDH80	1525	0	0	0	4	152	398	518	257	176	20	0	0
Wind	WSAvg	6.2	7.9	8.0	7.7	7.9	6.8	5.4	3.9	3.5	4.1	5.5	6.5	7.2
Precipitation	PrecAvg	34.6	1.9	1.6	2.2	3.1	4.4	4.0	3.7	3.9	2.8	2.8	2.5	1.6
	PrecMax	43.4	3.3	5.2	4.1	5.4	8.0	9.4	7.2	10.1	6.1	6.6	5.4	3.0
	PrecMin	21.7	0.5	0.5	0.9	1.5	0.8	1.0	1.0	0.6	0.4	0.2	0.2	0.4
	PrecStd	5.0	0.8	1.1	1.1	1.3	2.2	2.2	1.7	1.9	1.7	1.8	1.5	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.0	59.0	76.0	80.3	89.1	91.2	93.1	89.2	89.0	83.2	68.1	58.0
		MCWB	49.9	51.5	63.2	61.9	69.0	72.8	77.4	74.2	72.6	70.2	57.4	55.7
	2%	DB	45.7	49.6	65.1	75.2	84.3	87.9	88.2	85.7	85.5	77.2	61.6	49.9
		MCWB	42.6	45.4	56.9	59.7	67.4	72.5	76.1	74.3	71.5	62.8	53.2	46.3
	5%	DB	39.9	42.2	59.4	70.2	80.2	84.9	85.3	82.9	81.1	71.6	57.2	44.9
		MCWB	37.4	39.0	52.5	56.8	64.9	70.7	73.9	72.3	69.0	61.0	51.7	42.3
	10%	DB	36.3	37.8	53.7	65.4	75.9	81.5	82.7	80.3	77.7	67.1	52.9	40.6
		MCWB	34.5	35.3	47.8	54.1	63.5	69.6	72.2	70.4	67.1	59.4	47.2	38.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.3	54.1	64.3	65.2	72.9	77.8	80.1	78.1	76.0	69.8	60.2	56.8
		MCDB	52.2	58.2	75.7	74.0	84.4	86.1	88.7	85.2	85.0	77.3	65.0	58.6
	2%	WB	41.9	47.8	58.8	61.2	70.3	75.0	78.1	75.8	73.6	65.8	55.3	48.5
		MCDB	45.0	52.5	65.1	70.8	81.1	83.4	86.5	82.4	82.4	70.1	60.2	50.6
	5%	WB	36.9	40.6	54.1	58.3	68.1	73.2	76.2	73.9	71.3	62.7	51.4	43.0
		MCDB	39.0	43.4	58.9	66.9	78.2	81.9	84.0	80.8	79.0	68.7	56.9	45.5
	10%	WB	34.3	35.6	47.6	55.2	66.0	71.3	74.4	72.0	68.8	59.7	47.4	38.8
		MCDB	36.0	38.0	52.9	63.2	74.3	79.2	81.5	78.3	75.0	66.1	51.7	41.1
Mean Daily Temperature Range	5% DB	MDBR	14.2	15.0	16.8	21.5	20.9	19.7	19.8	20.2	23.0	20.7	16.8	13.4
		MCDBR	16.3	19.5	24.0	29.4	25.7	23.1	21.2	22.2	25.5	26.7	22.7	17.4
		MCWBR	14.6	15.7	16.1	17.1	12.7	11.9	11.6	12.8	11.7	14.3	15.9	15.0
	5% WB	MCDBR	15.6	20.9	22.1	24.9	21.3	19.6	19.0	19.4	19.9	20.5	20.9	17.3
		MCWBR	13.7	16.5	15.3	16.9	11.7	11.6	11.6	11.9	10.7	15.0	15.9	15.3
Clear-Sky Solar Irradiance	taub	0.285	0.303	0.334	0.387	0.415	0.424	0.437	0.421	0.394	0.342	0.301	0.283	
	taud	2.416	2.380	2.403	2.265	2.238	2.257	2.208	2.248	2.299	2.449	2.509	2.449	
	Ebn at Noon	275	287	290	280	273	271	265	266	265	269	269	266	
	Edh at Noon	25	30	33	41	43	43	44	42	37	28	23	22	
All-Sky Solar Radiation	RadAvg	533	830	1154	1476	1702	1931	1927	1697	1420	933	603	443	
	RadStd	60	104	98	122	126	143	109	101	121	109	52	48	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page